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Issues in future autonomous public transport solutions for children with intellectual disabilities

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Abstract: Public transportation in urban environments should be inclusive and accessible to offer this service for all people. This research aims to identify issues in the development of autonomous shuttle services in future public transportation systems for children with intellectual disabilities. An issue-based information system (IBIS) analysis was retrospectively conducted based on a concept generation phase. Four concepts were evaluated by five experts. The findings show that the main tensions in this context can be categorized as interaction, physical, and independence. Interactive elements can both create value and lead to confusion for the target group. The physical design of concepts affects feasibility, viability, and perception. Independence is a desired quality for the target group but often comes at a price of less independence for other stakeholders such as family members. Finally, the study suggests that there are tensions between the qualities of integrity, cognition, and user experience.

Keywords: Autonomous vehicles; Children with mild intellectual disability; Issue-based information system (IBIS); Design space analysis

1. Introduction

Artificial Intelligence (AI) and other technological advances have potential to rapidly change our urban environments. Autonomous vehicles (AV) often serve as prominent examples of AI-human interaction in daily life. They illustrate AI's autonomy and its integration into our physical and social environment. While there is extensive engineering research on autonomous vehicles, practical applications of such physical-digital technologies remain unexplored. These innovations in society bring changes that are both challenges and opportunities to inclusion and accessibility. It is essential that technological advancements are suitable for individuals with diverse needs and abilities. Various design approaches, including universal design, inclusive design, and design for all, have emerged to address concerns of accessi-

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bility (Holloway & Barbareshi, 2022; Persson et al., 2015). In this study we focus on the development of new autonomous shuttle services in a medium-sized Swedish city, and how they can support children with intellectual disability (ID).

Approximately 1.3 billion people live with some form of disability, making them the largest minority on Earth according to the WHO (WHO, 2023). Societal barriers faced by these individuals result from design choices—how we plan and construct our physical, social, and digital communities—choices that limit the participation of people with various impairments in daily activities. This is described in the Preamble of the Convention on the Rights of Persons with Disabilities (United Nations, 2006): "disability results from the interaction between persons with impairments and attitudinal and environmental barriers that hinder their full and effective participation in society on an equal basis with others".

Still, individuals with ID often experience a lack of control and choice when it comes to transportation (Bodde & Seo, 2009; Verdonschot et al., 2009), and they find that their specific needs, goals, and sense of control in daily life are not adequately met (Gjermestad et al., 2017). One significant area where this lack is felt is transportation, which plays a crucial role in people's daily lives. In Sweden, individuals with disabilities exhibit distinct mobility patterns. They rely proportionally more on cars for their travels compared to public transportation when compared to the rest of society, and their overall travel rates are lower (Trafikanalys, 2018).

Transportation services using AVs for children mean additional concerns, not the least among parents. Numerous studies have delved into the perspectives of parents regarding such services (e.g., Lee & Mirman, 2018; Lee et al., 2020; Tremoulet et al., 2020). The findings from these studies indicate that, particularly among women, parents have expressed apprehensions about the safety of their children in AVs. However, in some cases, they could also identify certain safety benefits and conveniences (Lee & Mirman, 2018; Tremoulet et al., 2020). In surveys, parents who indicated their reluctance to use AVs for transporting their children cited concerns related to child safety. These concerns included worries about potential harm, uncertainties about who would meet the children at the destination, and the need for security features such as onboard cameras and microphones to enable assistance when needed (Lee et al., 2020).

In an interview study of opinions about AVs among children, Larsson (2021) found that there is potential in AVs supporting independence for children. However, an element to consider is how children perceive and understand AVs, to be able to safely use services associated with them. Charisi et al. (2017) have looked at children's views and understandings about how AVs communicate their intentions. They emphasize the importance of designing these interactions in a comprehensible way to ensure safety. In conclusion, there is a pressing need for more comprehensive efforts to ensure that innovative technologies aimed at improving public transportation are truly inclusive (McDonnell et al., 2021).

The aim of this research is to identify issues in the development of autonomous shuttle services in future public transportation systems for children with intellectual disabilities. We

also explore the use of an issue-based information system (IBIS) approach for understanding a design space and identifying arguments, tensions, and further issues within this domain.

2. Design spaces

In this research project we are interested in finding ways of understanding the issues for introduction of AVs in public transport. We aim to break apart the reasoning behind a concept development phase and assume that it holds valuable information about how alternatives and solutions together form a web of critical issues to understand. We consider these issues as part of a design space and suggest that tensions between arguments specifically are interesting when designing for new situations and technologies.

Design space is an abstract construct of all the possible and identified ideas relevant to a design. To be able to understand the forces within a design space one can examine alternatives within that space. All designers create, discover, or uncover alternatives as they work. According to Akın (2001), a main distinguishing mark of expert designers compared with non-experts is the creation of comparatively more alternative problem formulations and not only alternative solutions to one or a few problem formulations.

There is a supposition that more ideas lead to better designs, and thus that more of the design space is explored, and that in addition to better design outcome, designers become better at justification of ideas and concepts with more exploration (Burgess Yakemovic & Conklin, 1990). Being able to share those ideas, and the thinking behind them, is important for collaboration and group decision-making. The idea is that making the thinking around a problem or within a design space available for scrutiny for all members to improve choices. We suggest that this approach can also lay bare the underlying forces, and how they relate to different choices, within a given design space. When examining a design space, understanding not only what choices have been made but also *why* they have been made is clearly helpful. This type of information is often lost in large projects or is difficult to summarize in a digestible way (ibid.).

An explicit account of a design space can be understood as a design rationale, that is, an articulation of the reasoning behind a solution and its features. The term design rationale is used both in a loose, informal sense corresponding to the abstract relationship behind a design and the choices made, and in a more formal sense as the physical documentation of the process. Although the exact components of a rationale differ between researchers, the original idea was introduced by Kunz and Rittel (1970). In this approach, design is viewed as a series of discussions around wicked problems (Rittel & Webber, 1973/1984). The problems are wicked in the sense that they do not have a final, definite answer and must be understood within its underdefined and perspective-dependent context. For this type of problem, each solution is part of a complex space of solutions influenced by subjective perspectives.

The recognition that the problems (or issues) we deal with in design have subjective components and complex structures motivates creating or analyzing the rationale behind design solutions (Moran & Carroll, 1996).

2.1. Issue-based information systems

IBIS offers a way to analyze design rationale using a kind of argumentation analysis, which is natural given that the original area of application for IBIS was political decision processes (Kunz & Rittel, 1970). The main part of IBIS is *issues*. The issues are the questions discovered during a process. Each issue has one or several *positions* that address the issue. And then a set of *arguments* that either support or oppose the positions.

Many automated approaches to generating a design rationale have been put forth. IBIS has been applied as a web-based tool, called IBISMod enabling distributed teams working together on shared problems to collaborate (Ebadi, Purvis, & Purvis, 2009). The Design Rationale editor (DRed), was also based on IBIS as a digital repository that makes it possible for the participants to work together on a specific problem while they may be geographically distributed during development (Kim, Bracewell, & Wallace, 2005). Ebadi, Purvis and Purvis (2009) offer further examples.

Making the design space visible and shareable is the main benefit of digital tools like these. One challenge for these systems is that it takes time and energy to document a thought process. Thus, getting people to use a system like this during development is a challenge, even though the potential benefits are documented. Though there is some evidence indicating that digital tools are becoming better at providing the needed answers during development processes (see e.g., Aurisicchioa & Bracewell, 2013).

Another, associated idea behind computerized tools in design space exploration is that design performance can be amplified by providing suggestions or automated processes for generating alternatives. Such tools in the digital domain generate variations based on rules such as e.g., cloning or motif (Woodbury, 1993). These rule-based approaches suit generative ability on the level of visual or compositional variation. With the recent developments in generative AI, there are increasing possibilities to enhance exploration on conceptual level using generative AI (Zhu & Lou, 2022; Bilgram & Laarmann, 2023; Tholander & Jonsson, 2023).

2.2. Design space analysis

A related approach to IBIS is design space analysis using the Question, Option, Criteria notation (QOC), which was proposed by MacLean et al. (1991). A QOC analysis is conducted as an additional step after design, analyzing the outcome by applying a predefined structure of questions, options, and criteria. MacLean et al. (1991) suggested that a key differences between the QOC approach and IBIS was that a QOC focuses on criteria for alternatives (while IBIS only include pros and cons), and that approaches like the computerized IBIS approaches mentioned earlier, include any issues that emerge during development, while QOC options specifically concern design choices that structure a design space. However, they also suggest ways in which the two approaches can be combined.

Both IBIS and QOC start in a problem (issue or question), but Westerlund (2005) argue that it is the artifact that does not yet exist that is in focus during the work, and it is the possible artifacts that make the design space. However, as Dorst and Cross (2001) successfully argue, design entails a co-evolution of a problem-solution space. A design rationale needs to address both problems and solutions.

In a study conducted by Lee (1997), a generic structure for design rationale systems was identified through a comprehensive survey of existing systems. According to Lee, the design rationale structure comprises three layers: 1) the *design intent* layer, 2) *decision* layer, and 3) *design artifact* layer. The design intent layer 1), positioned at the highest level, encompasses information guiding decisions throughout the design process. This layer represents intents, strategies, goals, and requirements driving the design decisions.

The decision layer 2), which follows, contains detailed information or rationale pertaining to the decisions made during the design process. This layer consists of five sub-layers that delineate the decision-making structure. These sub-layers include issues, arguments, alternatives, evaluations, and criteria. The issue layer comprises individual issues and their relationships, such as how they generate, depend on, or replace one another. The argument layer articulates the underlying arguments supporting the decisions, along with their relationships like support, refute, or qualify. The alternative layer outlines individual alternatives and their relationships, indicating how they are components of, incompatible with, or specialized versions of one another. Within the evaluation layer, the evaluation measures and the relationships between alternatives and these measures are expressed. The criteria layer specifies the criteria used to evaluate the alternatives and can be formulated based on information from the design intent layer. Lastly, the design artifact layer 3) illustrates the relationships between design components and can also link these components to the underlying rationale present in the decision layer.

Botero, Kommonen, and Marttila (2010) argue that design space analysis should include not only designers' moves during concepting, but also other co-designers and users who actively co-create and explore the design space during usage, which has its own set of constraint on their framing. Design space analysis can make designers, and design researchers, more conscious of limitations imposed by specific design decisions, how certain design activities have influenced the exploration, and what opportunities may have been missed (Dove, Hansen, & Halskov, 2016).

3. Research context and procedure

Autonomous vehicles, either as part of or parallel to existing public transportation, are currently being explored in several parts of Sweden. In Linköping, VTI, in collaboration with seven other entities, is involved in a project called Ride the Future. This project includes two autonomous buses operating in an area around Linköping University and the residential area

of Vallastaden. Ride the Future investigates how these vehicles can integrate into the existing mobility system and the potential they hold for enhancing travel options for people with various disabilities in a collective and adapted manner.

In this study we focus on the implementation and use of the Ride the Future shuttles and how they support autonomy of children with ID with the goal of generating concepts for future services. We used IBIS as a retrospective analysis tool to make sense of the concept generation phase of the project.

The project participants had earlier engaged the target group, children with mild ID, in riding the bus and conducting interviews. 16 children were included (M_{age}=9.75, SD_{age}=1.61) that had mild forms of ID. Ten of the children (M_{age}=9.9, SD_{age}=1.91) completed Raven's Progressive Matrices 2 (RPM 2, Raven et al., 2018) and scored equivalent to that of children aged 6.03 years (SD=1.19).The *project team*, consisting of 5 researchers (2 behavioral scientists, 2 cognitive science and service design researchers, and 1 information systems design researcher) acted as mentors for a *student group* consisting of 7 design and product development students working on their undergraduate thesis project.

The start of the project had an overarching question of how travels with self-driving buses can be designed for increased autonomy of children and adolescents with ID, and five key issues were identified by the project team based on the design research phase (Forsblad et al., 2023):

- Safety belts could be difficult to use.
- Safety belts for wheelchairs were complicated.
- Difficult to on- and off board using wheelchair.
- Difficult for the children to understand what the buses do, see, and communicate.
- Difficult to know where to sit.

The fourth element of the list became the focus of the team's design project. The complete process of going from project framing to final concept can be seen in Figure 1.

Prior research in the project had also revealed that familiar situations for the children were characterized by contemplation and curiosity, while unfamiliar ones were characterized by surprise or confusion, which lead to a question of how to design an explainable interior of a self-driving bus that takes retrospective, current, and prospective aspects into account (Arvola et al., 2023). The starting question for the research project became: How can the interior be shaped to be self-explanatory? In parallel, the student group conducted their own research phase and identified a goal for their project: The passenger wants to know before the bus arrives where to place themselves and exit the bus at the right time.

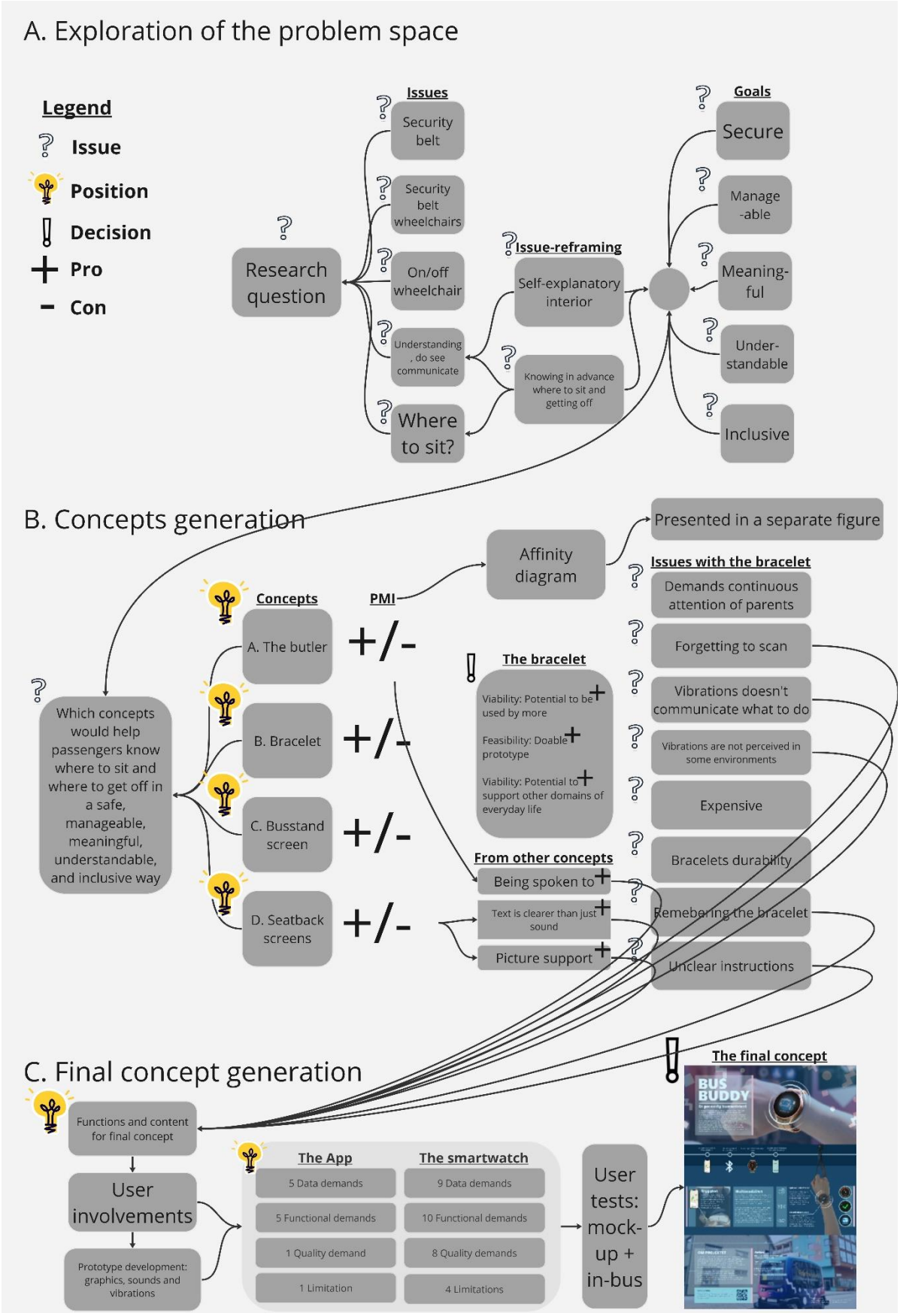


Figure 1 Process overview.

This led to a problem framing session where several aspects of a working solution were explored. The project team, based on the research and formulations of both the researchers and the students, then formulated a list of those aspects, see below. We use the word aspects here since the list items can be understood in diverse ways. One way to interpret them is to see them as requirements, but in practice they worked more as targets or goals for concept generation.

- Safe
- Manageable
- Meaningful
- Comprehendible
- Inclusive

A crucial element to note here is that it does not make sense to attempt to "solve" each of these issues separately. Instead, several different concepts were generated to further explore positions and find arguments. By going through those steps, the main issue to use as a starting point for concept generation was: What overall design solutions (concepts) could help passengers know where to sit and where to disembark in a safe, manageable, meaningful, comprehensible, and inclusive manner?

With that issue, formulated as a question, the student group generated four concept suggestions, with supervision from one member of the project team (Arvola). The project team was then invited to participate in a concept scoring/analysis where plus, minus, interesting (PMI) was used to evaluate the concepts. A PMI means you note positive feedback, negative feedback, and what you find particularly interesting about an idea (De Bono, 1982/1984). All team members filled in their feedback for the concepts using a template emailed to the participating researchers. The feedback was then the basis for a final concept suggestion developed from Concept B which consisted of an armband and associated app where parents of the children with ID could track and aid in planning travels (Figure 3). However, the actual analysis of the design space in this article takes all positive and negative arguments across all four concepts into account when identifying issues in the development and implementation of autonomous shuttles in future public transportation systems for children with intellectual disabilities.

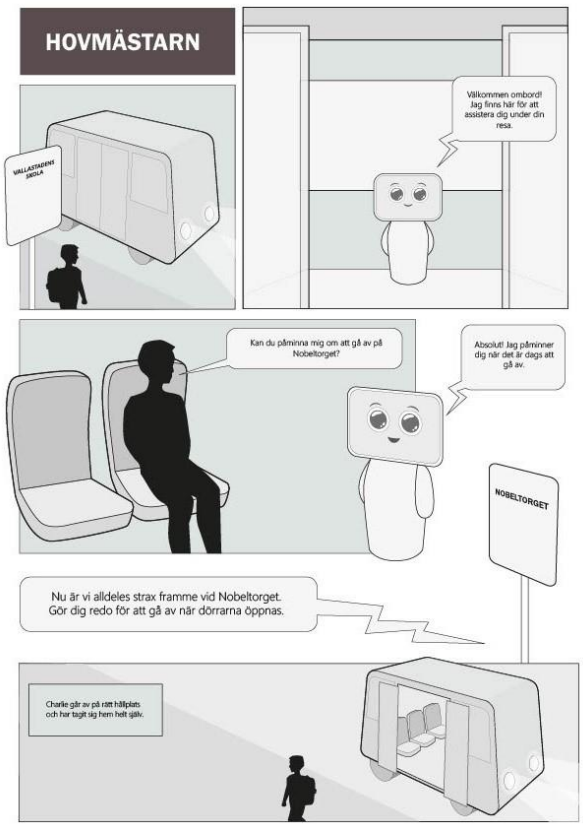


Figure 2 Concept A – The Butler

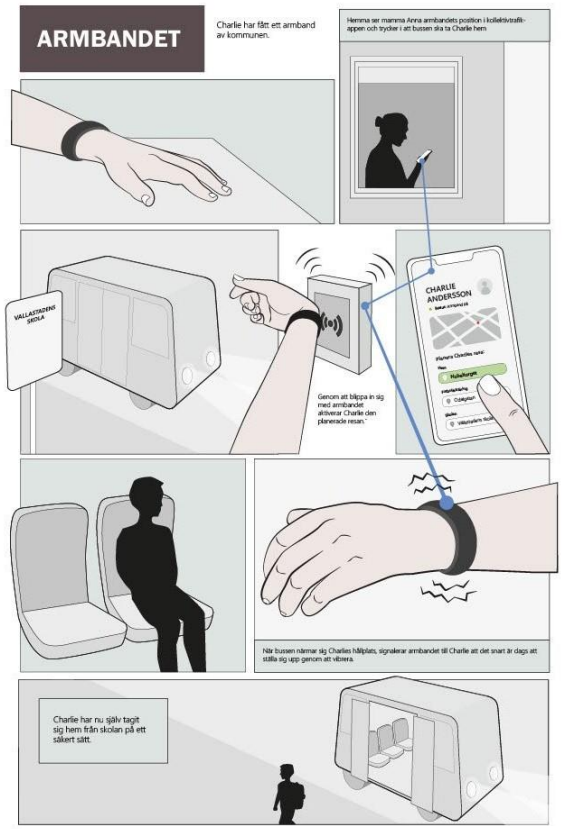


Figure 3 Concept B – The Bracelet

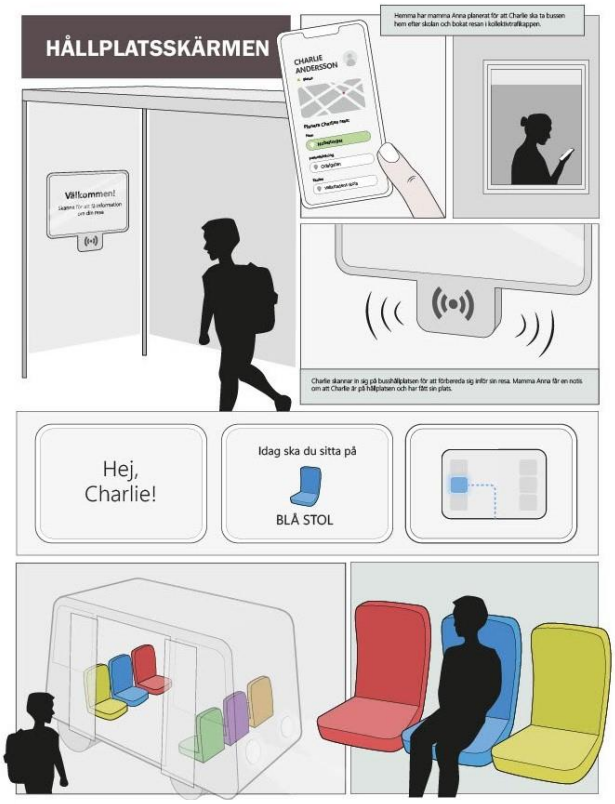


Figure 4 Concept C – The Bus Stand Screen

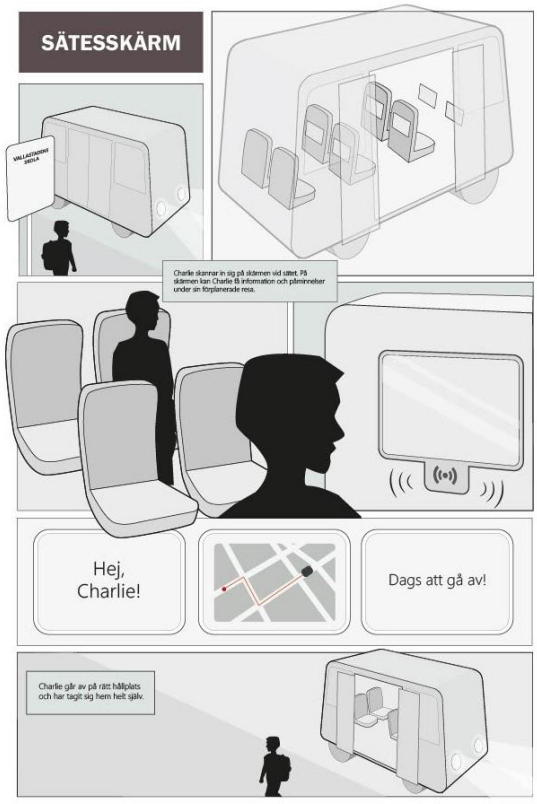


Figure 5 Concept D – The Seatback Screens

3.1. Analysis

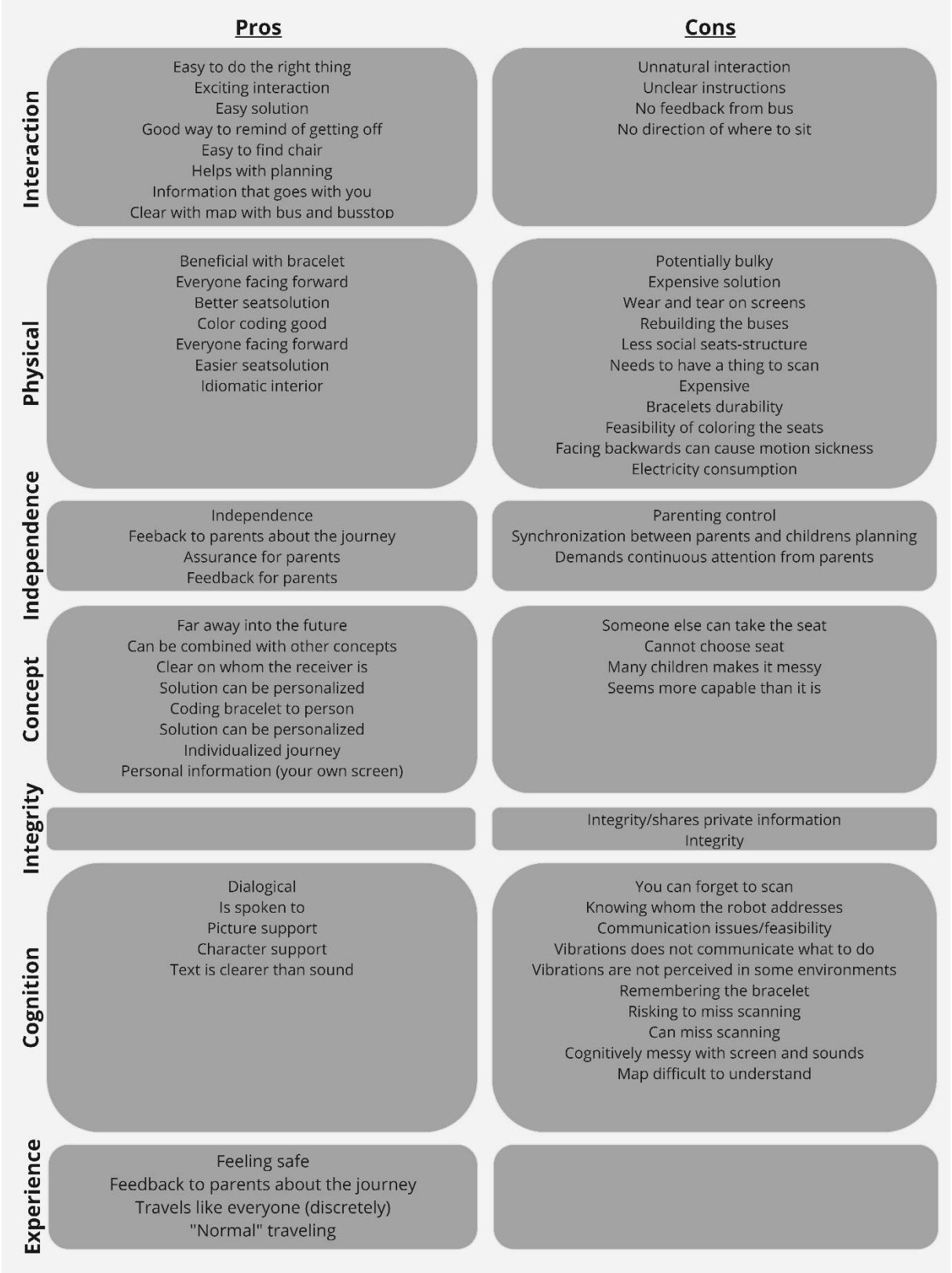


Figure 6 Affinity diagram over PMI Pros, Cons and Interesting. Interesting have been sorted under either Pros or Cons.

The analysis took the arguments generated during the PMI as input data. The arguments were documented using the concept templates sent to each project team member so extracting them was just a process of summarizing all feedback for each concept. The feedback provided as part of the "interesting" category was assigned to either the PRO or CON side depending on the comment. This was done by all the project team members collectively and by consensus. The next step was an affinity diagram where the content was organized according to overarching themes, in the positive and negative groups, respectively (see Figure 6). Overall, the process of identifying a starting issue and then conducting the analysis was done within 5 hours of work, split into two sessions.

4. IBIS outcome

The final themes of the analysis ended up being:

Interaction: The interaction theme contained positive arguments about making it easy to take correct actions, to find a seat and relevant information, ways to support traveling and making interaction exciting. On the negative side we find unnatural interaction, unclear instructions, no feedback or help with traveling. Overall, the interaction theme had more positive elements.

Physical: this category concerns the physical product, either something that the users can carry with them or the bus itself. The physical theme had more negative elements. The positive comments concerned ways in which wearables can be a helpful solution due to being always present, as well as comments about the interior of the bus, how seats were indicated with color and their placement in the shuttle. Concepts were considered negative if they were expensive, prone to break, or needed for system interactions (e.g., scanning something on board the bus). Also, regarding the interior, placement, and direction of seats, as well as color coding was seen as negative.

Independence: this theme is balanced between the positive and negative categories. On the positive side we have the perceived autonomy of the user and feedback/safety for the parents. On the other side, we have parent control, synching between parent and children and the need to be digitally available for parents. Hence, there is a tension here between independence, perceived autonomy of the child, and the control and effort required from the parent. More on that later.

Concept: this theme concerns arguments tied to the unique design concepts rather than to any other category. For instance, positive comments were made about a concept being futuristic, being easy to combine with other ideas, personalizing equipment, tailor-made trips, and individual information. Negative comments were made about concepts where children could have their designated chair taken by someone else, scenarios where they are unable to pick their own seat, environmental factors as well as solutions that seem more capable than they are.

Integrity: only negative comments are found in this theme. These comments have to do with how concepts deal with personal information and whether the children can travel just like everyone else.

Cognition: We also included communication problems in this theme. Positive feedback: dialogical, symbol-supported communication, and multimodality. Reported arguments against were that the children could forget to scan their device, be confused about who is being addressed by a robot/AI, using vibrations that are ambiguous compared to symbols or speech, mental or cognitive load and busy environments.

Experience: this theme only had positive associated comments and relates to how the target group is expected to experience the concepts. For instance, a feeling of safety, feedback for parents and traveling like everyone else were appreciated features.

5. Discussion

The IBIS could be used in this case to retrospectively break down the reasoning about concepts based on the PMI. The main tensions are found in the themes interaction, physical and independence.

The interaction supported by the final concept can influence the user experience of the travel in many ways. For instance, supporting travel by helping with travel plans, knowing what to expect and when to do something has immense potential value. Interactive elements can also add to the ability for users to receive and comprehend information by providing it at the right moment where it makes sense in relation to the current situation. Interactive elements can both create value and lead to confusion for the target group. The concepts suggests that some elements need to be added to the existing system of public transport if it is going to be accessible for the target group. The analysis of the concepts highlights how important interactivity is for the understanding of how to use the service properly. The design space for wearable technologies seems especially appealing for supporting the target group.

In the physical category of the analysis, we see that solutions impact feasibility by raising issues such as wear and tear or durability, changing the interior or rebuilding whole buses to accommodate potential solutions. Questions of viability relate to cost, energy and elements of the physical layout that influence social relations. The physical category also relates to perceivable aspects of the solutions that can be either positive or negative. PROs and CONs in the physical category need to be identified early in development projects before changes to the physical environment become too expensive. Products as part of the solutions might cause problems since some might not be able to afford to use them, some solutions where the interior of the shuttle is changed might not work because of brand perception, and finally there is an issue of wear and tear. This creates a sustainability issue that goes against the idea of adding another technological artefact. The design of solutions where the user is the owner of a new product also risks affecting how they are perceived from groups outside

or influence their ability to travel like everyone else negatively. However, there is also a terrific opportunity for using wearable technology, since it can provide context-sensitive information that makes it easier to understand.

Independence and autonomy are valued in this design space, but there is a clear tradeoff between the child´s autonomy and integrity and the parents' control. This tension was also reported in previous studies looking at parent's attitudes towards children using AVs (Lee & Mirman, 2018; Tremoulet, et al., 2020). If the cognitive activity of planning a journey is shared between child and parents, then it requires parents to be available constantly, thus limiting their independence. This is in turn connected to the experience category, where solutions that increase autonomy can impact the feeling of safety. For instance, if a parent always gets notified when their child enters a bus it can provide a feeling of safety. Similarly, knowing that the parent is notified might make the child feel safer. However, what happens when the system fails and no notification is sent, or when the child forgets to scan their wearable artifact at the bus? This is further connected to the theme of integrity, where parents are aware at any given time where the child is, which severely limits perceived independence.

These results indicate that the relationship between parents and children is going to be pivotal to understand and develop in future transportation services including AVs, and not the least developing services that are perceived as safe for parents (Lee, et al., 2020) but without putting too much effort and responsibilities on them. Below you can find the summarized arguments for and against in the theme of Independence:

Pros:

- Safety for parents
- Feedback for parents
- Autonomy

Cons:

- Requires parents' constant attention
- Syncing parents' planning with the child's
- Parental control

Based on the analysis we believe it is important for the target group to be able to grasp what the service can do for them. This is in line with previous research looking into children's understanding and communication with AVs (Charisi et al., 2017). Consequently, solutions that hold a promise to aid, but cannot fulfill that promise, might have large negative consequences. The analysis conducted here makes areas of development clear, and individual solutions can be traced back to concept choices by referring to the IBIS.

Lundberg et al. (2021) have argued that giving up autonomy at one level, for instance the control over steering a car, can lead to more autonomy at another level, as for example deciding where to go. In this case we can observe a different kind of *displacement of autonomy.* If the children gain independence and autonomy to go by themselves with an autonomous shuttle, then the parents can get less autonomy since they feel a need to check how it is going on their phones. Here we can see that autonomy is shifted between actors: children, vehicles, and parents.

5.1. Using IBIS to understand issues in design

This research does not use IBIS in a conventional way. Hence, it is not evaluating the usefulness of IBIS itself, but rather looking at its usefulness in discovering tensions in a design space by aiding in creating a rationale. Of course, the process is full of situated decision making, exploration and sense-making that are not part of the project's formal documentation. The only reason we can find the information that was used to make decisions along the way is that the students were obligated to send in a thorough report at the end of their course. Nonetheless, the approach was useful in identifying many tensions, issues, and arguments. As argued by Dove et al. (2016) the analysis forces us as researchers to examine how certain design activities have influenced the exploration, and what opportunities may have been missed (Dove, Hansen, & Halskov, 2016).

The specific approach reported here used information on the intent and decision layer of a design rationale, and less on the artefact layer (cf., Lee, 1997). We are combining the QOC approach with IBIS but since we are interested in the criteria and any information captured throughout the process. At the same time, we are building up the structure after the fact (and not during the design process) in line with how QOC is intended to be used.

After the IBIS analysis is conducted it is possible to examine the potential influence of individual features of concepts. By studying the themes and identifying tensions, and then tracing those back to individual concepts and solutions, it is possible to understand more about what e.g., an armband as a solution means for independence. By cross-refencing the concepts and the tensions, it is also possible to say something about alternatives, which are important for making sense of a design space (Burgess Yakemovic & Conklin, 1990).

6. Conclusion

In the research project reported in this paper we have explored inclusive public transportation for all people, with a particular focus on new technologies for children with cognitive disabilities. By conducting a retrospective analysis based on IBIS this research has identified issues in the development of autonomous shuttles in future public transportation systems for children with intellectual disabilities. The key issues found were related to interaction, physical, and independence. Independence was closely tied to integrity, which in turn is related to both cognition (aiding the children with cognitive functions while preserving integrity) and user experience (parents overview can be reassuring but also lead to a feeling of being monitored).

We argue that the possibilities brought by innovative technologies, and AVs specifically need to be met by inclusive and universal design approaches (Holloway & Barbareshi, 2022; Persson et al., 2015). This way we open solutions that are accessible and inclusive, not just for specific target groups but for people in general.

We believe this approach to breaking apart the design space and identifying the main tensions for future technologies might be a good approach, to complement inclusive design approaches. The value of this kind of analysis lies not in the specific solution and the forces (positions and arguments) leading to that design, but rather in the knowledge generated by looking at aggregated argument data

A limitation of this study is that the design space was limited to the four concepts that focused on what the passengers wants and need to know before the bus arrives, where to seat themselves, and where exit the bus at the right time. This is a valuable exploration, but future research should explore the design space of other parts of the service. Such exploration should include how the service for a target group such as children with intellectual disabilities can mitigate issues of understanding what the buses do, see, and communicate.

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